

WASP-12b

R-0292

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_240216 · created 2026-07-17T18:41:56+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460355.70814 +/- 0.00057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.0035
Transit depth (Rp/Rs)^2	2.4 +/- 0.11 [%]
Orbital Inclination (inc)	86.78 +/- 1.03
Airmass coefficient 1 (a1)	1.433 +/- 0.051
Airmass coefficient 2 (a2)	-0.3178 +/- 0.0077
Scatter in the residuals of the lightcurve fit is	3.39 %
Best Comparison Star	None
Optimal Aperture	3.8
Optimal Annulus	4.75
Transit Duration (day)	0.13393 +/- 0.00094

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.0035 vs 0.1178 (deviation 2.82σ)
Duration fit vs published	0.13393 d vs 0.125 d (deviation 2.52σ)
Mid-time O–C	-1575.4 min
Depth SNR	11.8
Residual scatter	3.39 %

Light curve

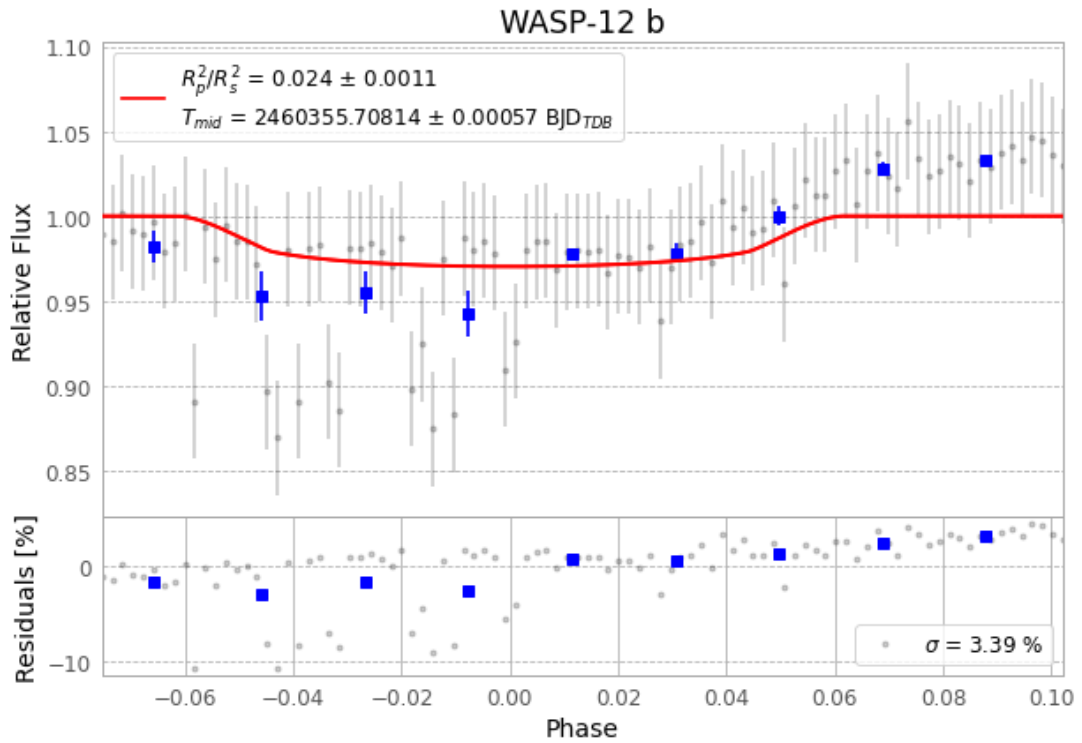


Figure 1 — FinalLightCurve_WASP-12 b_2024-02-14.png (SHA-256 04bd515612cecd2c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 248], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0eadc930747e0c8a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a5addef

Data provenance

94 FITS frames (62 MB), WASP-12240215025416.FITS → WASP-12240215073315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-240216.log (da9def632145...), FinalLightCurve_WASP-12 b_2024-02-14.png (04bd515612ce...), FinalLightCurve_WASP-12 b_2024-02-14.csv (e9cb5c5b4862...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 18:41Z.